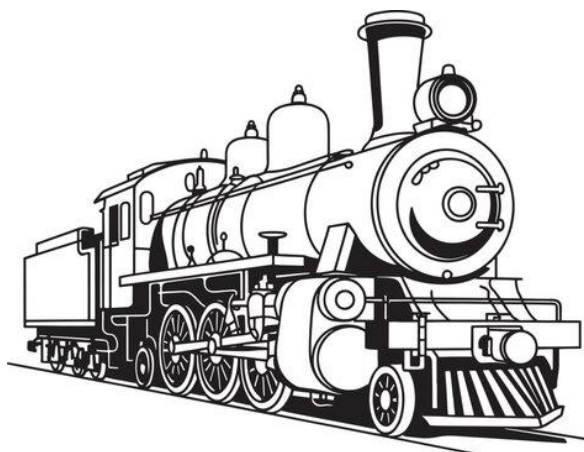


2025 UT Austin Invitational Tournament
October 25, 2025



DISEASE DETECTIVES

Division C Exam

Anaya Bawiskar, Biomedical Engineering, '27 (anabaw@tamu.edu)

Suhani Goswami, Computer Science & Biology, '27 (suhanigoswami@utexas.edu)

TEAM NAME:

TEAM NUMBER:

PARTICIPANT NAMES:

Instructions:

1. You should have this exam packet and a blank answer sheet. All answers must be recorded on the answer sheet to be graded. Anything written on this exam will **not be graded**. Be mindful of which section you are writing your answers in!
2. You will have 50 minutes to complete this exam.
3. A non-programmable, non-graphing calculator is permitted for this exam. Round all answers to the nearest hundredth.
4. You may take apart the documents, but all documents must be stapled together at the end.
5. There are a total of **240 points** on this exam.
6. In the case of a tie, the team with a greater score on Case Study #1 and Case Study #2 will win the tie.

Ever since you were 4, working at the CDC has been your dream career.
 Yesterday, you received a call that could make that dream come true: an interview for a
field epidemiologist position!

Vocabulary (1 pt each, 25 total)

Match the words in the table with their definitions below.

Write the LETTER on the answer sheet.

A: Sporadic	B: Endemic	C: Virulence	D: Random Error	E: Primary Case	F: Area Map
G: Environment	H: Systematic Error	I: Placebo Effect	J: Spot Map	K: Iatrogenic	L: Secondary Case
M: Hyperendemic	N: Cluster	O: Herd Immunity	P: Double-Blind	Q: Confounding Error	R: Morbidity
S: Pathogenicity	T: Mortality	U: Fomite	V: Index Case	W: Host	X: Infectivity
Y: Agent	Z: Outbreak	AB: Vector	AC: Epidemic	AD: Nosocomial	AE: Pandemic

1. A disease that occurs infrequently and irregularly.
2. The constant presence of a disease or infectious agent within a given geographic area or population group; may also refer to the usual prevalence of a given disease within such area or group.
3. A disease that is constantly present at a high incidence and/or prevalence rate.
4. The occurrence of more cases of disease than expected in a given area or among a specific group of people over a particular period of time.
5. An aggregation of cases of a disease or other health-related condition, particularly cancer and birth defects, which are closely grouped in time and place.
6. An epidemic occurring over a very wide area (several countries or continents) and usually affecting a large proportion of the population.
7. A beneficial health outcome resulting from a person's anticipation that an intervention will help.
8. A type of clinical trial in which neither the participants nor the researcher knows which treatment or intervention participants are receiving until the clinical trial is over
9. An inanimate object that may be contaminated with microorganisms and serves in their transmission.
10. Relating to an illness acquired in a hospital.
11. Relating to illness caused by medical examination or treatment.

12. A factor, such as a microorganism, chemical substance, or form of radiation, whose presence, excessive presence, or (in deficiency diseases) relative absence is essential for the occurrence of a disease.
13. An extrinsic factor (geology, climate, insects, sanitation, health services, etc.) which affects the agent and the opportunity for exposure.
14. A person or other living organism that can be infected by an infectious agent under natural conditions.
15. The first person who acquires the disease from exposure.
16. The first case brought to the attention of professionals.
17. Occurs when there is a difference between the true value and the observed value.
18. The result of fluctuations around a true value due to sampling variability and can occur with data collection, coding, transfer, or analysis of data.
19. Occurs when the effects of two risk factors are mixed in the occurrence of the health-related event under study.
20. A map that indicates the location of each case of a rare disease or outbreak by a place that is potentially relevant to the health event being investigated, such as where each case lived or worked.
21. The resistance of a group to invasion and spread of an infectious agent, based on the resistance to infection of a high proportion of individual members of the group.
22. Any departure, subjective or objective, from a state of physiological or psychological well-being.
23. The proportion of persons with clinical disease who, after becoming infected, become severely ill or die.
24. The proportion of persons infected, after exposure to a causative agent, who then develop clinical disease.
25. The proportion of persons exposed to a causative agent who become infected by an infectious disease.

Acronyms (2 pts each, 8 total)

Write the full form of the following acronyms:

1. NHANES:
2. CDC:
3. SHEA:
4. NCHS:

Epidemiological History (1 pt each, 14 total)

Match the following pandemics to the year they happened:

A: 1347 - 1351 B: 1817-1923 C: 1855 D: 2009
E: 2014 - 2016 F: 2002 - 2004 G: 1889 - 1890

1. Russian Flu:
2. Cholera:
3. Ebola:
4. SARS:
5. Bubonic Plague:
6. Swine Flu:
7. The Third Plague:

Choose the correct answer:

8. Who is the current director of the CDC?
 - a. Rochelle P. Walensky
 - b. Jim O'Neil
 - c. Susan Monarez
 - d. Robert R. Redfield
9. Who was the first person to publicly and successfully demonstrate the use of ether anesthesia for surgery?
 - a. Kiyoshi Shiga
 - b. Anton van Leeuwenhoek
 - c. John Graunt
 - d. William T. G. Morton
10. Who is the father of epidemiology?
 - a. John Graunt
 - b. John Dalton
 - c. John Snow
 - d. John Gordon
11. Which of the following was the first antibiotic created in 1928?
 - a. Amoxicillin
 - b. Oxacillin
 - c. Ampicillin
 - d. Penicillin

12. Who is considered the father of surveillance for his work in the collection and analysis of Britain's mortality data?
 - a. Louis Pasteur
 - b. John Graunt
 - c. William Farr
 - d. Joseph Goldberger
13. The first formal, large-scale human experiment was a trial for which vaccine?
 - a. Polio
 - b. Smallpox
 - c. Rabies
 - d. Typhoid
14. Which of the following diseases was the CDC established to combat the spread of?
 - a. Tuberculosis
 - b. Malaria
 - c. Polio
 - d. Salmonella

Koch's Postulates (3 pts each, 9 total)

1. Explain the significance of the postulate "The cultured microorganism should cause disease when introduced into a healthy organism" in 1-2 sentences.
2. A team of researchers is investigating the outbreak of a respiratory illness in a town. They find an unknown bacterium present in the individuals who have the disease, but do not find it in healthy individuals in the town. However, the scientists are unable to grow this bacterium in the lab. Write the complete postulate that is not being satisfied in this investigation.
3. An epidemiologist is studying Salmonella, related to an outbreak at a new restaurant in town. They were able to isolate the Salmonella from the food that was eaten by a majority of the diseased individuals. Write 2-3 sentences on the next steps the epidemiologist should take to fulfill the remaining postulates.

Surveillance (2 pts each, 18 total)

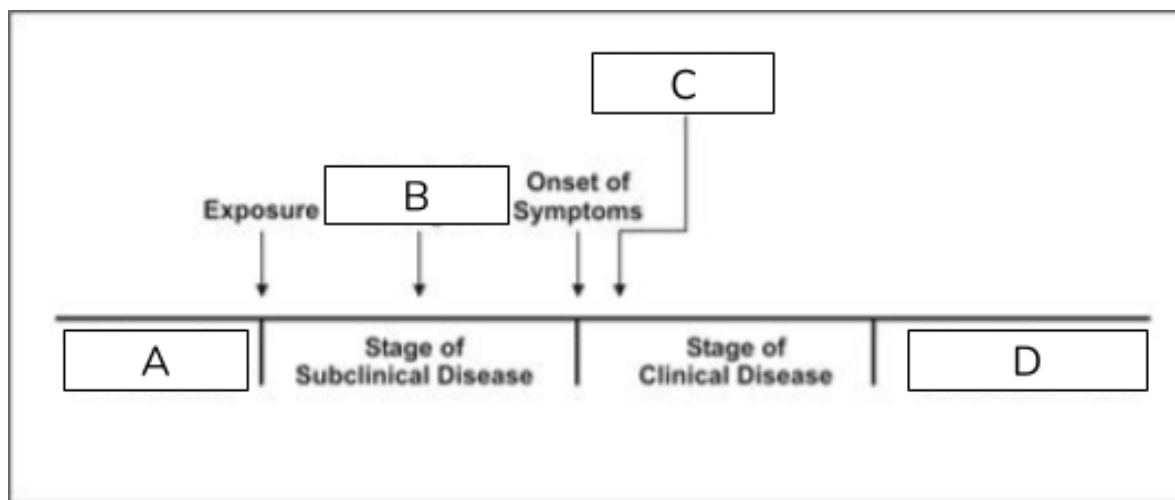
1. Write the steps in order in the Public Health Surveillance Process.
2. In 1-2 sentences, describe the purpose of Surveillance.

Identify the type of Surveillance being described in the following scenarios:

3. Following a hazardous chemical spill, workers in the affected area were monitored for 72 hours for any symptoms of chemical exposure.
4. A hospital's laboratory automatically forwards all positive test results for COVID-19 to the state health agency. This is part of the state's routine reporting requirements.
5. After identifying a case of a new, highly contagious flu strain, health department officials call all hospitals and clinics daily for one week to ask specifically about patients with similar symptoms.
6. A public health official receives an alert when a hospital's emergency room reports a spike in the number of patients reporting respiratory symptoms, a pattern that doesn't align with the usual seasonal flu trends.
7. A person who was in contact with a confirmed case of a communicable disease checks their temperature daily and reports any symptoms to a nurse for two weeks.
8. A group of large hospitals has been chosen by health officials to conduct annual surveys to analyze the prevalence of a disease in their areas. These hospitals provide data that is used for public health policies, even though they only represent a fraction of hospitals in their state.
9. A system that monitors pharmacy sales reports an unusual increase in the purchase of over-the-counter cough and cold medication, prompting an investigation into a potential outbreak.

Natural History of Disease (1 pt each, 4 total)

Fill in the missing parts of the timeline:



Hill's Criteria (2 pts each, 30 total)

Scenario	Hill's Criteria
1. The association is replicated in different settings and using different methods.	A. Strength
2. The cause and effect have a 1:1 relationship.	B. Coherence
3. An increasing amount of exposure to the agent increases disease risk.	C. Analogy
4. Exposure always precedes effect.	D. Consistency
5. Similar relationships have been observed in other contexts.	E. Specificity
6. The relationship is consistent with biological principles.	F. Plausibility
7. The condition can be prevented or induced by controlling conditions.	G. Experiment
8. The relationship is compatible with current theory and practice.	H. Biological Gradient
9. There is high correlation between variables.	J. Temporality

Match the scenarios below with what aspect of Hill's Criteria they are describing using the letters from the table above.

10. People who smoke cigarettes more often are significantly more likely to develop cancer.
11. The death rate from lung cancer rises linearly with the number of cigarettes smoked daily.
12. A retrospective and prospective study on smoking produce similar results.
13. Mesothelioma is almost exclusively caused by asbestos exposure, and rarely by any other factor.
14. Lung cancer develops later in life for someone who smoked for years as a teenager.
15. Carcinogens in tobacco smoke can damage DNA, causing mutations that may lead to cancer.

Bias (2 pts each, 18 total)

Name the specific type of bias described in the following scenarios:

1. In a 10-year weight loss trial, participants who fail to lose weight are more likely to drop out.
2. A study comparing factory workers to the general population finds workers have lower overall mortality, partly because only relatively healthy people can maintain employment.
3. A treatment appears more effective overall, but when results are broken down by age group, the treatment is less effective in both younger and older patients.
4. Parents of children with leukemia are more likely to remember pesticide exposure in the home than parents of healthy children.
5. When asked about alcohol consumption in a survey, college students underreport drinking because they want to appear more responsible.
6. A study finds an association between coffee drinking and heart disease, but the true risk factor is smoking, which is more common among coffee drinkers.
7. A cancer screening test appears to improve survival time, but patients only seem to live longer because their disease is diagnosed earlier, not because they actually survive longer.
8. A cancer screening program detects more slow-growing tumors than fast-growing ones, making it seem as if screened patients have better outcomes, even if screening does not improve mortality.
9. Researchers conduct a study on exercise habits and heart disease but recruit participants only from a local fitness club.

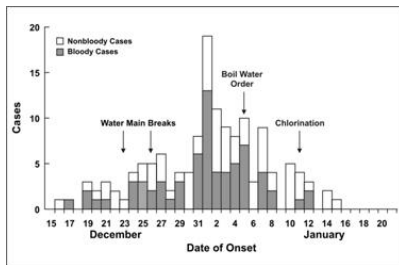
Epidemic Patterns (2 pts each, 18 total)

Name the specific epidemic pattern described by each of the following statements.

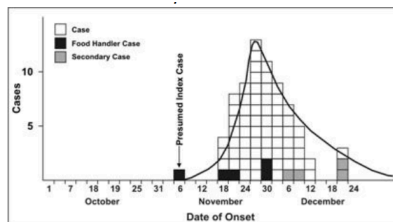
1. A group of persons are all exposed to an infectious agent or a toxin from the same source.
2. A group is exposed over a relatively brief period and becomes ill within one incubation period.
3. Patients are exposed to an infectious agent over a period of time, resulting in a flat and wide epi-curve.
4. Transmission between patients occurs, resulting in more than one incubation period with multiple peaks on an epi-curve.
5. The level of exposure to an agent naturally fluctuates, resulting in multiple, irregular waves in cases on an epi-curve.
6. A pattern that has characteristics of both common-source and propagated epidemics.

Name the epidemic pattern shown in each epi-curve.

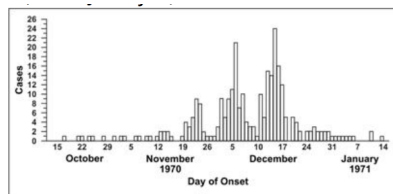
7.



8.



9.



Types of Studies (2 pts each, 12 total)

Name the type of study being described.

1. Randomly assigns participants to an experimental group or a control group.
2. Enrollment into the study is based on the presence or absence of disease.
3. Looks backwards and examines exposures to suspected risk or protection factors in relation to an outcome that is established at the start of the study.
4. Follows over time groups of individuals who are alike in many ways but differ by a certain characteristic.
5. Situations where nature is allowed to take its course.
6. A group of people is observed, or certain information is collected, at a single point in time or over a short period of time.

Screening Test (3 points each, 18 total)

178 people consume sashimi at a restaurant and are tested for a new disease called Anahani. 49 people have positive results, and of these, 11 are false positives. Out of the negative results, 102 are true negatives.

Based on this information, answer the following questions.

1. Calculate the sensitivity and explain what these results mean in 1 sentence. Provide your answer as a percent.
2. Calculate the specificity and explain what these results mean in 1 sentence. Provide your answer as a percent.
3. Calculate the positive predictive value and explain what these results mean in 1 sentence. Provide your answer as a percent.
4. Calculate the negative predictive value and explain what these results mean in 1 sentence. Provide your answer as a percent.
5. Calculate the prevalence of disease. Provide your answer as a percent.
6. In 1 sentence each, define false positives and false negatives.

Case Study #1 (Points defined by each question, 36 total)

Remember that interview from the beginning of the test? You aced it! Your interviewer was impressed with your in-depth knowledge and attention to detail. The director wants to meet with you personally, so a train ticket from your hometown of Boston is sent to your email, and you depart early the next day for Washington DC.

You have a nice lunch on the train, watching the sights and enjoying the ride, drinking your matcha, listening to Clairo, your tote bag with your Labubu clipped on sitting on the seat beside you. :p

Mysteriously, around 2 hours after lunch at 2:30 pm on September 29th, 2025, fellow passengers started raising concerns about nausea, vomiting, headache, and dizziness! Oh no!

1. Write a case definition for this outbreak. (3 points)

You interview people about the foods they consume, including people who feel sick and those who don't. With all your information jotted in your notes app, you head back to your seat and create the following table.

Lunch Item	Sick	Not Sick
Burger	6	11
Salad	3	20
Pasta	16	15
Wrap	7	19
Baked Potato	3	13

2. What type of study is this? (1 point)
3. Name two benefits and two drawbacks of this type of study. (4 points)
4. Is calculating an odds ratio or a relative risk more appropriate for this study? Explain in 1 sentence. (2 points)
5. Based on your answer to number 3, calculate the appropriate measure for each food in the table. Show your work for full credit. (10 points)
6. Calculate the Chi-Square Test Statistic for the Burger and Pasta data. Show your work for full credit. (8 points)
7. What food has most likely caused this outbreak? Explain in 1 sentence. (1 point)
8. What type of bias most likely influenced this study? (1 point)
9. [Division C Only] Name 3 strategies to prevent future infections. (6 points)

Case Study #2 (Points defined by each question, 30 points total)

After the news of your great work on the train case, you have officially been hired as a field epidemiologist for the CDC! In fact, your first case starts tomorrow!

You travel to Austin, TX, to investigate reports from urgent care facilities of patients experiencing cough, fever, nausea, and congestion.

You review the profiles of 10 cases that checked into Bevo Urgent Care in the last month.

Label	Date of Onset	Age	Gender	Location of Residence	Symptoms	Lab Result
a.	09/30/2025	30	M	Round Rock, TX	Cough, Fever, Headache, Congestion, Vomiting	Negative
b.	10/12/2025	43	F	Round Rock, TX	Cough, Fever, Nausea, Congestion	Pending
c.	10/12/2025	23	F	Austin, TX	Cough, Fever, Nausea, Congestion, Hearing Loss	Positive
d.	10/13/2025	26	M	Buda, TX	Cough, Nausea, Cramps, Fever, Headache, Hearing Loss	Pending
e.	10/15/2025	56	F	Austin, TX	Fever, Nausea, Diarrhea	Negative
f.	10/16/2025	34	M	Pflugerville, TX	Cough, Fever, Constipation	N/A
g.	10/20/2025	19	M	Round Rock, TX	Cough, Fever,	N/A

				TX	Nausea, Headache, Congestion	
h.	10/21/2025	31	M	Buda, TX	Cough, Fever, Nausea, Congestion, Hearing Loss	Positive
i.	10/22/2025	22	F	Austin, TX	Cough, Fever, Nausea, Headache, Loss of Taste	Pending
j.	10/22/2025	47	F	Manor, TX	Fever, Headache, Fatigue	N/A

1. What type of chart is shown above? (1 point)
2. What information would be needed to consider this an outbreak? (2 points)
3. What type of surveillance was used? (2 points)
4. Write a case definition for this scenario. (3 points)
5. What is the next step for an outbreak investigation, assuming the previous steps were completed? Include both the step number and name. (2 points)
6. Label each case as Confirmed, Probable, Suspected, or Not a Case. (0.5 points per question, 5 pts total)

After receiving test results from many of the cases, you find that the outbreak resulted from Hookem virus, a disease that is transmitted by close proximity to infected people who are coughing, sneezing, or talking.

7. What is the most likely mode of transmission of Hookem virus? (2 points)
8. Define the Basic Reproduction Number (R_0). (2 points)
9. Define herd immunity. (2 points)
10. Assume a Basic Reproduction Number of 4. What is the herd immunity threshold? Express your answer as a decimal to two places. (3 points).

11. If the proportion of the population that is immune is 0.50, what is the effective reproductive number? (3 points)
12. Assuming that vaccination is 90% effective, calculate the adjusted herd immunity threshold. Express your answer as a decimal to two places. (3 points)

Congratulations on finishing the test! Happy Halloween :D